Hexadecimal

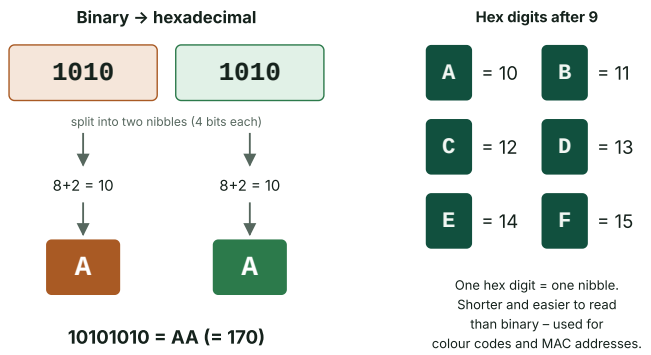
D04 FACT FILE

Specification: students should be able to

- Demonstrate understanding of and use number representation and convert between denary, binary and hexadecimal

CCEA fact file: Digital Data

Key words: hexadecimal, base 16, nibble, hex digit, colour code, MAC address



Converting the byte 10101010 to hexadecimal AA by splitting it into nibbles

Hexadecimal

Hex is base 16, using the digits 0–9 and the letters A–F for 10–15. Because 16 = 2^4, one hex digit represents exactly one nibble, so any byte can be written as two hex digits. Programmers use hex because it is much shorter than binary, easier to read and less error-prone, and it converts directly to binary. Common uses include colour codes, MAC addresses and memory addresses.

Conversions

Binary to hex: split into nibbles and convert each to a hex digit. Hex to binary: write each digit as 4 bits. Denary to hex: divide by 16 – the quotient is the first digit and the remainder the second (or go via binary). Hex to denary: multiply the first digit by 16 and add the second.

T Key terms from the mark schemes

Hexadecimal
Base-16 number system (0–9, A–F); a shorthand for binary.
Credited: U4 2019 Q8, 2023 Q7

Exam focus

Hex conversion appeared in 2019 Q8(b)–(c) (including uses of hex), 2022 Q6(b)(ii), 2023 Q7(c) (66 to hex and an advantage of hex) and 2024 Q4(b)(ii) and (c)(ii). Show the nibble split for working marks.

Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q7	[13]	Bit, 47 to binary, 66 to hex, advantage of hex, binary addition and overflow
Unit 4 2022 Q6	[14]	Binary digit, 54 to binary, nibble, hex, overflow, storage capacity
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Convert the binary number 10101010 into hexadecimal. Show your working. [2]
2. Convert the denary number 66 into hexadecimal. Show your working. [3]
3. Convert the hexadecimal number 3B into denary. [2]

Answers: 1. Split into nibbles: 1010 | 1010 = 10 | 10 (1) / AA (1) 2. Either method: 66 ÷ 16 = 4 remainder 2 (1) / the quotient 4 is the first digit and the remainder 2 the second (1) / 42 (1). Accept via binary: 66 = 01000010 (1) → 0100 | 0010 (1) → 42 (1). 3. 3 × 16 + 11 (B = 11) = 48 + 11 (1) / = 59 (1)